

Health Memory — An AI-Agent-Powered Persistent Health Memory Layer for Elderly Patients

Keeping an elderly patient's health story connected, understandable and securely available to everyone who cares for them.

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01 Title of the Project

Health Memory — An AI-Agent-Powered Persistent Health Memory Layer for Elderly Patients, Guardians, Doctors, Caregivers, Old-Age Homes and Pharmacists.

02 Problem Statement

An elderly patient's health information is scattered across doctors, pharmacies, caregivers and hospitals that never communicate with one another — so a changed dose, an illegibly handwritten prescription or a known allergy is lost at exactly the moment it matters most, leading to avoidable medication errors, repeated tests and unsafe emergency care.

India has over 149 million citizens aged 60 and above. A typical elderly patient consults a specialist at one hospital, a physician at a neighbourhood clinic, buys medicines from a pharmacy that keeps no record, and is cared for daily by a family member or facility attendant who has never met either doctor. The information exists; the connection between the people holding it does not.

03 Solution Proposed

A working web platform on which an elderly patient or their guardian builds a single, permanent, **consent-controlled health memory** from three kinds of everyday input, and then decides — scope by scope — which of the people caring for them may read which part of it. Four role-specific AI assistants read that memory under those permissions and answer with visible evidence.

The core pipeline

01 Add Type it, say it, or photograph the document — including handwritten prescriptions.	02 Understand OCR, handwriting recognition and medical entity extraction produce structured facts, each with a confidence score.	03 Verify Low-confidence <i>and</i> clinically important fields are routed to a licensed pharmacist for human confirmation.	04 Connect Authorised doctors, caregivers and pharmacists retrieve only what consent permits, with the source shown.
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Key features

FEATURE	WHAT IT DOES
Three-mode capture	Text, voice (browser speech recognition with a typed fallback) and scan/upload. All three enter the same pipeline and keep their original wording.
Handwriting OCR	A dedicated recognition engine for handwritten prescriptions, with calibrated confidence and approximate drug-name matching against a medical lexicon.

FEATURE	WHAT IT DOES
Confidence-gated verification	Every extracted field is scored. A field reaches a pharmacist only if it is below the trust threshold <i>and</i> clinically important.
Consent Control Center	The patient grants or revokes seven information scopes across five recipient types. Every change is logged and takes effect immediately.
Doctor routing	A scanned prescription that names a prescriber is automatically associated with that doctor — without ever fabricating a doctor account.
Four role dashboards	Patient/Guardian, Doctor, Caregiver & Old-Age Home, Pharmacist — one record, four working surfaces.
Shift-based care workflow	Medication rounds, care tasks, observations and an AI-assisted handover for individual caregivers and facility rotas.
One-click visit summary	A longitudinal, evidence-backed draft the doctor reviews and edits before it is saved — never auto-committed.
Emergency Health Card	Blood group, allergies, critical conditions, verified medications, emergency contact and primary doctor — readable at arm's length and printable.
Four role-specific agents	Separate assistants for each role, each answering in the appropriate register, each showing its sources.

Technology

LAYER	TECHNOLOGY	RATIONALE
Backend	Python, FastAPI, SQLAlchemy	Typed contracts and automatic API documentation for review and audit.
Database	SQLite (pilot) → PostgreSQL	Zero-setup pilots; a one-line change to a production engine.
Document AI	Tesseract OCR with a separate handwriting path	Open-source, offline-capable; the engine interface accepts a trained handwriting model later.
Retrieval	Consent-filtered RAG with local hashed embeddings	Runs offline with no API key; character-shingle hashing tolerates OCR noise.
AI layer	Provider abstraction: deterministic grounded provider or hosted LLM	The system answers correctly with no model at all; the LLM only improves phrasing.
Frontend	React 18 + Vite, responsive, accessibility-oriented	Larger type scale and plain language on the patient surface; usable on a caregiver's phone.
Security	PBKDF2 hashing, signed session tokens, role and patient-level authorization, audit log	Standard-library cryptography; no unvetted dependency in the auth path.

Mechanisms that make it trustworthy

- **Provenance on every fact.** Each record permanently carries its source, author, confidence and verification status. Patient-reported information is never presented as a doctor-confirmed diagnosis.
- **Selective human-in-the-loop.** A doctor's name misread at 66% does not interrupt anyone; a dose misread at 66% does. The routing rule is written once and applies everywhere.
- **Consent enforced before retrieval.** Unauthorised records are removed from the candidate set before relevance is ever scored — the language model is never handed something it is asked to ignore.
- **Non-destructive correction.** A pharmacist's correction is stored beside the original reading and its score, never on top of it.

Current status

This is not a concept. A complete working prototype exists: a full backend, four functioning dashboards, the OCR and verification pipeline, consent-aware retrieval and all four assistants, verified by an automated end-to-end test suite.

31	60	4	95	0
database tables in the health-memory schema	working API endpoints	role dashboards and AI agents	automated checks passing, zero failures	external API keys required to run

04 Implementation Plan

Target beneficiaries

BENEFICIARY	SCALE IN THE PILOT	DIRECT BENEFIT
Elderly patients & guardians	60–100 residents / home-care patients	One place holding their whole health story; no repeated explanation of history at every visit.
Old-age home caregivers	8–15 staff across three shifts	Structured medication rounds and handovers; a refusal mechanism that prevents administering unverified medication.
Treating doctors	4–8 consultants and physicians	Longitudinal context before the consultation begins; a visit summary drafted in one click.
Pharmacists	2–3 verifying pharmacists	A focused queue of only the readings that genuinely need a human eye.
Family members	Guardians of each patient	Visibility and control over who sees what, from anywhere.

Deployment phases

PHASE	DURATION	ACTIVITY	EXIT CRITERION
1 • Harden	Weeks 1–4	Encryption at rest and key management; PostgreSQL migration; expand the drug lexicon to a full formulary; role-based onboarding flows.	Security review passed; no plaintext clinical data at rest.
2 • Field trial	Weeks 5–10	Single old-age home in Chennai. Staff and caregiver training; digitise existing paper records; run parallel with the paper register.	60+ patient records live; caregivers completing shifts in-app without reverting to paper.
3 • Clinical loop	Weeks 11–16	Onboard the facility's visiting doctors and a partner pharmacy. Activate doctor routing, visit summaries and the verification queue end to end.	Verification queue cleared within one working day; doctors using summaries in real consultations.
4 • Evaluate	Weeks 17–20	Measure against the baseline captured in Phase 2. Structured interviews with patients, guardians, caregivers, doctors and pharmacists.	Evaluation report with measured outcomes against the targets below.
5 • Extend	Weeks 21+	Second and third facility; home-care patients with individual caregivers; regional language interface.	Three deployments running without on-site engineering support.

Expected outcomes

These are pilot targets to be measured, not claims. Each is defined against a baseline captured during the first two weeks of Phase 2, using the facility's existing paper register as the comparison.

OUTCOME	TARGET	HOW IT IS MEASURED
Medication administration errors	Reduction vs. baseline	Facility incident register, before and after, over equal periods.
Time to prepare a shift handover	Under 5 minutes	Timed observation of handover preparation, paper vs. platform.
Doctor preparation time before consultation	Reduction vs. baseline	Timed from opening the record to being ready to consult.

OUTCOME	TARGET	HOW IT IS MEASURED
Illegible-prescription queries to prescriber	Reduction vs. baseline	Pharmacy call log for clarification requests.
Emergency information availability	100% of patients	Proportion of residents with a complete Emergency Health Card.
Caregiver adoption	Sustained daily use	Shifts started and handovers saved per week, by staff member.

Deliverables

TECHNICAL

Production-hardened platform with encryption at rest · deployment and operations guide · complete API documentation · automated regression suite · data migration tooling for existing paper records.

OPERATIONAL

Role-specific training material for caregivers, doctors and pharmacists · onboarding checklist for a new facility · consent explainer for patients and families in plain language and regional language.

EVIDENCE

Baseline and post-deployment evaluation report · anonymised usage analytics · incident and near-miss analysis · structured user feedback from all five beneficiary groups.

ACADEMIC

Publishable study on confidence-gated human verification in clinical document processing · open reference implementation of the consent-aware retrieval architecture.

05 Safety and Efficiency Analysis

Scope note. The materials this system handles are *medication, medical documents and clinical information*. The analysis below addresses safety and efficiency in the handling of those materials and in the daily care tasks built around them.

Safety — five mechanisms, each enforced in the backend

RISK	MECHANISM	WHY IT HOLDS
A misread dose or drug name is administered	Confidence-gated pharmacist verification, then a hard administration interlock	A caregiver attempting to record an unverified medication as administered receives a server-side refusal, not a dismissible warning. The rule cannot be bypassed from the interface.
An AI answer is mistaken for a clinical fact	Provenance carried inline into every retrieved record; every AI output labelled and shown with its sources	The assistant physically cannot present an unverified reading as confirmed without contradicting the labelled text in front of it. Every answer exposes a "why am I seeing this?" evidence panel.
An AI summary silently becomes the medical record	Draft-and-review: generation and acceptance are separate, audited steps	A visit summary is stored as a draft until a clinician edits and saves it; only then is it recorded, and it is stamped as a doctor-reviewed AI draft.
Information reaches someone it should not	Patient-level authorization plus consent filtering applied before retrieval	A signed-in doctor with no care relationship is refused and the refusal is logged. Unauthorised records never enter the candidate set, so no prompt instruction is relied upon.
An error cannot be traced after the fact	Non-destructive correction plus an append-only audit log of 21 action types	The original reading, its confidence, the correction, the pharmacist and the timestamp all persist together. Every access, retrieval, consent change and verification is recorded.

What the system refuses to do

- It does not invent a medication, dose, diagnosis, date, result or doctor.
- It does not fabricate a doctor account — an unregistered prescriber is recorded as an external provider so the source survives.

- It does not prescribe, alter treatment, or present OCR uncertainty as fact.
- It does not list unverified medication on the Emergency Health Card.

Efficiency — where the time goes today, and where it goes after

TASK	CURRENT HANDLING	WITH THE PLATFORM
Retrieving a patient's history	Searching a paper file or asking the family to recall it; often re-taken from scratch.	A longitudinal timeline retrieved in seconds, with the source of every entry.
Reading a handwritten prescription	Pharmacist telephones the prescriber, or interprets it under time pressure.	Only genuinely uncertain, clinically important fields are queued — with the original page, the reading and its score side by side.
Shift handover	Verbal, or a handwritten register; detail is lost between shifts.	A structured handover assembled from the shift's actual medication records, observations and pending tasks, reviewed before saving.
Medication rounds	A paper chart per resident; missed doses noticed late or not at all.	Rounds generated automatically for the shift window; missed doses surface immediately as alerts.
Preparing for a consultation	Reconstructing what changed since the last visit from whatever the patient brought.	A one-click draft covering history, medications, changes, labs, caregiver observations, alerts and unresolved issues.
Emergency response	Searching for records, or proceeding without them.	A single card with blood group, allergies, critical conditions and verified medications — printable and readable at arm's length.

Efficiency by design, not by exhortation

- **Selective escalation.** Because only clinically important low-confidence fields queue, a pharmacist reviews a handful of items per document rather than every field — the difference between a workable queue and an ignored one.
- **Capture where the patient already is.** Voice input and phone-camera capture mean no data-entry step is added to anyone's day.
- **Write once, read everywhere.** A caregiver's observation is simultaneously the doctor's context and the next shift's handover material. No duplicate recording.
- **Zero external dependency.** The system runs fully offline with no API key, so a connectivity failure in a facility does not stop care.

06 Scalability and Future Development

Technical scalability

DIMENSION	TODAY	PATH TO SCALE
Data layer	SQLite, single file	PostgreSQL via a single connection-string change; no application code depends on SQLite.
API layer	Single FastAPI process	Stateless, token-authenticated requests — horizontally scalable behind a load balancer without session affinity.
Retrieval	Local hashed embeddings, per-patient corpus	Retrieval is already patient-scoped, so cost grows per patient rather than with the whole corpus. A vector database or hosted embedding model is a one-function substitution.
Document processing	Synchronous OCR on upload	Move to a queue-backed worker pool; the ingestion service is already a discrete, independently callable unit.
AI layer	Deterministic provider or hosted LLM	Provider abstraction accepts any model, including a locally hosted one for data-residency requirements.
Configuration	Every threshold in one file, environment-overridable	Per-deployment tuning without a code change or a rebuild.

Organisational scaling

- **One facility** → **a chain**. The facility, staff, shift and assignment model already exists; multi-facility isolation is a tenancy boundary on top of it.
- **Institutional** → **district**. Consent scopes and audit logging are the preconditions for a public-health deployment where data crosses institutional lines.
- **National integration**. The layer is designed to federate with India's ABDM / ABHA health ID rather than compete with it — a patient's ABHA number becomes their identity, and this becomes the consent and provenance layer on top.

Roadmap, in dependency order

#	ENHANCEMENT	WHY IT COMES AT THIS POINT
1	Encryption at rest and key management	Blocks everything else for real patient data.
2	Full drug formulary with interaction checking	Replaces the pilot lexicon; enables a new safety class — flagging dangerous combinations, not just misreadings.
3	Trained handwriting recognition model	Drops into the existing engine interface; directly reduces pharmacist workload.
4	Per-field threshold calibration from an annotated corpus	Needs real pilot data to fit; turns reasoned defaults into measured ones.
5	Regional language interface and voice	Tamil first; the largest single accessibility gain for the target population.
6	Time-limited and break-glass consent	Emergency access with automatic expiry and mandatory post-hoc review.
7	Native mobile with offline capture	For caregivers in low-connectivity facilities; sync on reconnection.
8	ABDM / ABHA federation	National interoperability once the platform is proven in the field.

Adaptation to adjacent environments

The architecture is not specific to geriatrics. The same memory layer — provenance, confidence, verification, consent — transfers directly to:

POST-OPERATIVE CARE

Recovery at home with a rotating set of visiting nurses, where handover and medication tracking are the entire problem.

CHRONIC DISEASE

Diabetes and hypertension management across a physician, a specialist and a pharmacy that share no record.

RURAL PRIMARY CARE

A PHC where the visiting doctor changes and offline operation is mandatory, not optional.

PALLIATIVE CARE

Where family, nursing staff and a palliative team must stay aligned on a rapidly changing plan.

MATERNAL HEALTH

Antenatal records that must follow the mother across ANM, PHC and hospital.

DISABILITY SUPPORT

Long-term care where a support worker's daily observation is clinically meaningful.

07 Innovation Advantage

Four things that do not exist together in any current product

1 · Handwriting is a first-class input, not a failure case. Most document pipelines treat a handwritten prescription as a bad scan to be rejected. Here it has its own recognition engine, its own confidence calibration and its own downstream consequence. Because OCR of cursive script returns fragments rather than clean words, drug extraction matches *approximately* against a medical lexicon and **lowers its own confidence by the similarity gap** — reporting "probably Amlodipine, but only 54% certain" rather than silently asserting it. The raw reading is preserved for the human reviewer.

2 · Uncertainty is routed to a human, selectively. A verification task is created only when a field is *both* below the trust threshold *and* clinically important. A doctor's name read at 66% confidence does not interrupt anyone. A dose read at 66% does. This is the difference between a verification queue a pharmacist actually works and one they learn to ignore — and it is what makes human-in-the-loop economically viable at scale.

3 · Consent is a retrieval filter, not an interface toggle. Unauthorised records are removed from the candidate set *before* relevance is scored, so the AI is never handed information it is merely instructed to ignore. Revoking a scope removes those records from the timeline, the summary and the assistant in the same instant. Most systems enforce permissions at the screen; this one enforces them at the data.

4 · Provenance survives every transformation. A pharmacist correcting a misread drug name does not overwrite anything. The original OCR string, its confidence, the corrected value, the pharmacist's identity and the timestamp all persist together and remain visible on the record months later. Trust is a property of the data, not a claim in the marketing.

Comparison with existing approaches

EXISTING APPROACH	WHAT IT DOES WELL	WHERE IT LEAVES THE ELDERLY PATIENT
Hospital EHR / HIS	Comprehensive within one institution; clinically mature.	The record belongs to the institution. An elderly patient's care does not happen inside one institution, so the record fragments the moment they leave.
Personal health record apps	Patient-owned; simple document storage.	A folder of photographs. No structure, no verification, no way for a doctor to receive only the relevant part, no trust model.
ABDM / ABHA health ID	National identity and consent framework; the right foundation.	Provides linkage, not comprehension. It does not read a handwritten prescription, score it, or route uncertainty to a pharmacist. This platform is designed to sit on top of it, not against it.
Medical chatbots / LLM assistants	Fluent natural-language interaction.	Answer from a model, not from a verified record. No provenance, no consent boundary, no human verification — and they fail unsafely rather than visibly.
Old-age home management software	Rosters, billing, basic medication charts.	Operational, not clinical. No connection to the patient's doctors, no document understanding, no shared memory beyond the facility walls.

The defensible position. Competitors are strong on either *connectivity* (health IDs, EHRs) or *comprehension* (document AI, LLM assistants). This project is built at the intersection, and adds the piece both are missing: a calibrated, auditable account of *how much each fact should be trusted*, with a human in the loop exactly where — and only where — it matters.

Stated honestly. The prototype uses a pilot-scale drug lexicon and reasoned rather than corpus-fitted thresholds, and does not yet encrypt data at rest. These are named in the roadmap as prerequisites, not omitted from it.

08 Visual Impact & User Journey

Persona: Radha, 72. Hypertension and type 2 diabetes. Lives in an old-age home in Chennai. Sees a cardiologist at one hospital and a physician at a nearby clinic. Her son manages her care from another city.

Before and after — the same five moments in Radha's care

STAGE	1 · SYMPTOM APPEARS	2 · DOCTOR VISIT	3 · NEW PRESCRIPTION	4 · DAILY MEDICATION	5 · EMERGENCY
BEFORE	Dizziness for days × Mentioned to a caregiver, written in a register nobody else reads. Forgotten by the appointment.	History reconstructed from memory × The doctor asks what she is taking. Nobody is certain. A test is repeated because the last result cannot be found.	A handwritten slip × Illegible. The pharmacist guesses or telephones the clinic. The dose change never reaches the facility.	Paper chart per resident × A missed dose is noticed days later. The night shift does not know what the morning shift observed.	Nothing to hand × The paramedic does not know her blood group, her allergy, or what she takes. Treatment begins blind.
AFTER	Recorded in her own voice ✓ Spoken in 20 seconds, stored as patient-reported, timestamped, and already in the doctor's context.	Longitudinal record, already open ✓ What changed since the last visit, current medications, caregiver observations and the last lab result — in one click.	Photographed, read, scored, verified ✓ Uncertain drug and dose go to a pharmacist, who confirms them. The verified change reaches the facility the same day.	Shift rounds and handover ✓ Doses generated for the shift window; a missed dose raises an alert; the handover carries every observation forward.	Emergency Health Card ✓ Blood group, allergy, critical conditions, verified medications and contacts — on a phone or printed, readable at arm's length.

The moment that changes everything

Stage 3 is where the project earns its place. A single handwritten prescription is the point at which an elderly patient's care most often goes wrong — and it is the point every existing product either ignores or fails at silently.

BEFORE — SILENT FAILURE

The slip says something like "Amlodipine 5 mg". The pharmacist cannot be sure. There is no time to call, so the most likely reading is dispensed.

The caregiver administers what is in the box. The facility register still shows the old dose. Nobody records that anyone was ever uncertain.

If harm follows, there is no trace of where the error entered.



AFTER — VISIBLE, RESOLVED, RECORDED

The page is photographed. The handwriting engine reads it and scores each field. "Amlodipine" comes back at 54% — below the trust threshold, and clinically important.

A high-priority task reaches a pharmacist with the original page beside the reading. She confirms it. The caregiver cannot administer it until she has.

The record permanently shows: read at 54%, confirmed by a named pharmacist, on a given date.

What changes for each person

PERSON	BEFORE	AFTER
Radha (patient)	Repeats her history at every visit; cannot say with confidence what she takes.	Her story is held for her, in her own words where she chose to speak it, and she decides who reads it.
Her son (guardian)	Manages her care by telephone, with no visibility.	Sees the record and controls consent from another city.
The caregiver	Works from a paper chart; inherits an incomplete verbal handover.	Structured rounds, immediate alerts, and a handover built from what actually happened in the shift.
The doctor	Reconstructs history from recall; repeats tests.	Opens a longitudinal record and a drafted summary before the consultation begins.

PERSON	BEFORE	AFTER
The pharmacist	Interprets illegible handwriting under time pressure.	Reviews only genuinely uncertain, clinically important readings, with the source page in view.

The single sentence that summarises the change. Before, the information existed but the connection did not — and nobody could say how much any particular fact should be trusted. After, the record follows the patient, carries its own evidence, and asks a human being for help at exactly the points where being wrong would matter.

Confidence figures quoted throughout are produced by the working prototype on its seeded demonstration documents and will vary with the recognition engine installed. All patient data used in development is fictional.